

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/ Supplementary Winter Examination – 2023

Course: B. Tech. **Branch :** Civil Engineering/Civil & Infra **Semester :** V

Subject Code & Name: Concrete Technology (BTCVC504/BTCIC504)

Max Marks: 60

Date: 08/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks														
Q. 1 Solve Any Two of the following.		12														
A) Define Cement. Explain any six types of Cement.	CO 1	6														
B) Explain in brief: Accelerator, Plasticizer & Super plasticizer.	CO 2	6														
C) Explain Bulking of Sand in detail.	CO 1	6														
Q.2 Solve Any Two of the following.		12														
A) Define Workability. Explain Slump Cone Test with diagram.	CO 3	6														
B) Illustrate Wet manufacturing process of Cement with flow chart.	CO 1	6														
C) Define Aggregate .Find Fineness Modulus. State its limits.	CO 1	6														
<table><tr><td>Sieve Size</td><td>4.75 mm</td><td>2.36 mm</td><td>1.18 mm</td><td>600 micron</td><td>300 micron</td><td>150 micron</td></tr><tr><td>Weight Retained</td><td>50 gm</td><td>300 gm</td><td>220 gm</td><td>240 gm</td><td>150</td><td>40 gm</td></tr></table>			Sieve Size	4.75 mm	2.36 mm	1.18 mm	600 micron	300 micron	150 micron	Weight Retained	50 gm	300 gm	220 gm	240 gm	150	40 gm
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Q. 3 Solve Any Two of the following.		12														
A) Define : a) Durability b) Creep c) Segregation d) Bleeding e) Fresh Concrete f) Green Concrete	CO 3	6														
B) Explain grades of Cement & grades of Concrete.	CO 1	6														
C) Explain difference between Tilting Mixer & Non Tilting Mixer.	CO 3	6														
Q.4 Solve Any Two of the following.		12														
A) Write a short note on AAR & Sulphate attack on concrete.	CO 3	6														
B) What is NDT? Explain any one method in detail.	CO 3	6														
C) Define Admixture. State its types & Uses.	CO 2	6														
Q. 5 Solve Any one of the following.		12														
A) a) Write difference between High Strength Concrete & High Performance Concrete?	CO 3	6														

b) Explain Duff Abraham's Law in detail.

CO 1 6

OR

B) Design a concrete mix for M45 grade using following data:

CO 3 12

- a) Grade designation : M45
- b) Type of cement : OPC 43 Grade (Specific Gravity=3.15)
- c) Maximum nominal size of aggregate : 20mm (Specific Gravity=2.80)
- d) Minimum cement content : 320 kg/m³
- e) Maximum water-cement ratio : 0.45
- f) Workability : 125 mm slump
- g) Exposure condition : Severe (RCC)
- h) Method of concrete placing : Pumping
- i) Volume of C.A. (20 mm) & for F.A. Zone II= 0.62 (W/C=0.50)
- j) Type of Aggregate: Crushed Angular Aggregates
- k) Super Plasticizer will be used.
- l) Fine aggregates: Confirming to Zone II (Specific Gravity=2.70)

Table 2 Assumed Standard Deviation
(Clause 4.2.1.3)

Sl No.	Grade of Concrete	Assumed Standard Deviation N/mm ²
(1)	(2)	(3)
i)	M10	3.5
	M15	
ii)	M20	4.0
	M25	
iii)	M30	5.0
	M35	
	M40	
	M45	
	M50	
	M55	
	M60	
iv)	M65	6.0
	M70	
	M75	
	M80	

Table 4 Water Content per Cubic Metre of
Concrete For Nominal Maximum Size of
Aggregate
(Clause 5.3)

Sl No.	Nominal Maximum Size of Aggregate mm	Water Content ¹⁾ kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

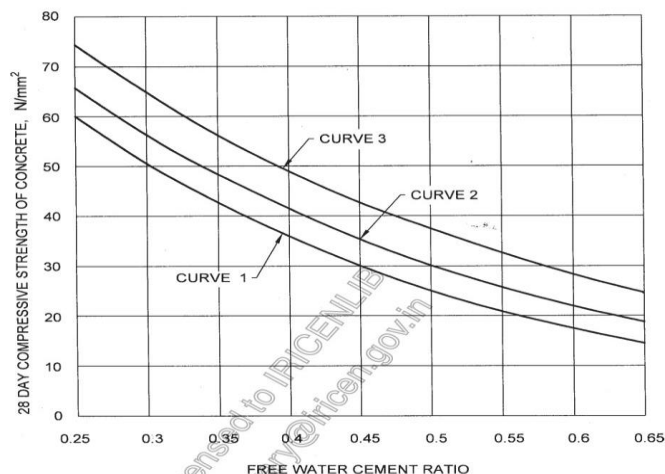
¹⁾ Water content corresponding to saturated surface dry aggregate.

NOTES

1 These quantities of mixing water are for use in computing cement/cementitious materials content for trial batches.

2 On account of long distances over which concrete needs to be carried from batching plant/RMC plant, the concrete mix is generally designed for a higher slump initially than the slump required at the time of placing. The initial slump value shall depend on the distance of transport and loss of slump with time.

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Curve 1 : for expected 28 days compressive strength of 33 and < 43 N/mm².
Curve 2 : for expected 28 days compressive strength of 43 and < 53 N/mm².
Curve 3 : for expected 28 days compressive strength of 53 N/mm² and above.

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